Terminal

```
PS C:\Users\cidco> docker run --rm --gpus all hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

PS C:\Users\cidco>
```

Figure 2: Confirmation of Docker installation

Terminal

```
Status: Downloaded newer image for nvidia/cuda:12.4.0-runtime-ubuntu22.04

==========
PS C:\Users\cidco>    docker run --rm --gpus all nvidia/cuda:12.4.0-runtime-ubuntu22.04 nvidia-smi

==========
== CUDA ==
==========

CUDA Version 12.4.0

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Tue Jul 29 09:52:42 2025
+---------------------------------------------------------------------------------------+
| NVIDIA-SMI 535.247.01             Driver Version: 535.247.01   CUDA Version: 12.4     |
|-----------------------------------------+----------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |         Memory-Usage | GPU-Util  Compute M. |
|                                         |                      |               MIG M. |
|=========================================+======================+======================|
|   0  NVIDIA L4                      Off | 00000000:31:00.0 Off |                    0 |
| N/A   43C    P0              29W /  72W |      0MiB / 23034MiB |      5%      Default |
|                                         |                      |                  N/A |
+-----------------------------------------+----------------------+----------------------+

+---------------------------------------------------------------------------------------+
| Processes:                                                                            |
```

Figure 3: Confirmation of NVIDIA process

needed for an AI application. It helps developers use Docker as a fully managed service.

Once your request to utilise Docker Offload is granted, you will see a screen as shown in Figure 1.

We can check GPU acceleration by starting a container with the *--gpus* option as shown below:

```
# docker run --rm --gpus all
hello-world
```

If you get an output as shown in Figure 2, Docker Offload is working fine.

To check the configuration of the NVIDIA process, we can use the command given below:

```
# docker run --rm --gpus all
nvidia/cuda:12.4.0-runtime-
ubuntu22.04 nvidia-smi
```

You can see the output in Figure 3.

The advantages of Docker Offload are:

- Remote build based on cloud.
- NVIDIA L4 GPU backend, which helps developers run any compute intensive workload seamlessly.
- Encrypted tunnel between the Docker desktop and cloud environment.

To know more about this fully managed service provided by Docker, you can go to *https://docs.docker.com/offload/about/*.

END

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